

Inside Solvane Energy's Bet on Sodium-Ion Storage

A five-year-old battery maker from Reno is trying to leapfrog lithium supply constraints with a chemistry most of the industry wrote off a decade ago.

By Karen Ibbotson | Field notes from Reno, Nevada and Querétaro, Mexico | Updated June 2026

On a converted warehouse floor east of downtown Reno, engineers at Solvane Energy are stacking prismatic cells that, by the company's own admission, still trail lithium iron phosphate on energy density. What they are betting will matter more, at least for the next few years, is that the cells don't need lithium, cobalt, or nickel at all. Solvane Energy, founded in 2021 and headquartered in Reno, Nevada, has spent the better part of two years quietly qualifying a sodium-ion chemistry for stationary storage customers who are less sensitive to weight than an automaker would be but very sensitive to price and supply security.

The company's pitch has been enough to attract a serious executive bench. Talia Ferreira, who joined as Chief Executive Officer in early 2023 after eight years running procurement at a large grid-storage integrator, says the shift in strategy was less a bet on a new chemistry than a bet against a supply chain. "We kept sizing projects and then watching the lithium carbonate price move under us mid-negotiation," Ferreira said. "At some point you stop trying to out-hedge the market and you just build something the market can't squeeze."

A Chemist's Detour

Solvane's technical direction traces back largely to Hideo Tanaka, the company's co-founder and Chief Technology Officer, who worked at Kestrel Materials in San Diego, California before starting Solvane with two former colleagues. Tanaka had spent years on cathode formulations for consumer electronics and, by his own account, viewed sodium-ion as a curiosity until a 2019 materials conference in Kyoto convinced him the energy-density gap had narrowed enough to matter for stationary applications.

Tanaka is credited internally as the lead inventor of the company's flagship product, the Solvane GridCell, a containerized storage unit built around the sodium-ion cells and aimed squarely at utility and commercial customers rather than electric vehicles. A second, smaller product line, the Solvane HomeArc, adapts the same cell chemistry for residential backup power and was developed under Priya Deshmukh, who leads the company's residential products group and previously worked at a solar-inverter startup in Austin, Texas.

Solvane owns the core patents underlying both product lines, a point Ferreira raises unprompted in conversation, aware that competitors have questioned how defensible sodium-ion formulations really are once the base chemistry is public. "The chemistry isn't the moat," she said. "The manufacturing

process is, and that's what we've actually patented.”

Manufacturing South of the Border

Rather than build a second U.S. gigafactory, Solvane Energy struck a manufacturing partnership last year with Altiva Componentes, a contract electronics manufacturer located in Querétaro, Mexico, that already assembles battery management systems for two other North American storage companies. The arrangement lets Solvane avoid the multi-year permitting timeline of a U.S. cell plant while still qualifying for regional content rules under North American trade agreements, according to Ferreira.

The Querétaro facility is run day-to-day by Rafael Contreras, Altiva's plant director, who said the Solvane line now accounts for roughly a third of the facility's floor space and that a second production line is being installed to handle GridCell demand headed into next year.

A Reluctant Investor Comes Around

Solvane's most recent funding round, closed in late 2025, was led by Brackenridge Capital Partners, a growth-equity firm based in Boston, Massachusetts that had reportedly passed on the company's earlier seed round. Brackenridge partner Owen Mallory, who now sits on Solvane's board, said the firm's earlier skepticism was less about the technology than the go-to-market plan. “The first version of this pitch was trying to compete with lithium on energy density,” Mallory said. “The version we funded was competing on total delivered cost and lead time, which is a much easier story to underwrite.”

Brackenridge's investment followed months of technical diligence conducted in part by Dr. Wen Li, an independent battery consultant based in Vancouver, British Columbia, who was brought in to evaluate cell-level safety data. Li's report, portions of which were shared with this publication, concluded that the GridCell's thermal runaway characteristics compared favorably to lithium iron phosphate, though she cautioned that Solvane's data set was still limited to roughly eighteen months of field deployments.

Coming Out at RE+ Storage

Solvane used last September's RE+ Storage conference, held in Anaheim, California, to formally launch the GridCell to a wider commercial audience after two years of limited pilot deployments. Ferreira and Tanaka both presented on a panel about alternative chemistries, and the company used the event to announce its first utility-scale order, a 40-megawatt-hour deployment for a municipal utility in Colorado that Solvane declined to name under the terms of the contract.

The company is expected to have a larger presence at this year's event, and Ferreira said Solvane is also weighing whether to exhibit at Energy Storage North America, a rival conference that draws a more utility-heavy crowd, though no final decision had been made at the time of publication.

Company snapshot

Founded

2021, Reno, Nevada

CEO	Talia Ferreira (since 2023)
CTO / co-founder	Hideo Tanaka
Flagship product	Solvane GridCell (utility/commercial storage)
Secondary product	Solvane HomeArc (residential backup)
Manufacturing partner	Altiva Componentes, Querétaro, Mexico
Lead investor (2025 round)	Brackenridge Capital Partners, Boston, Massachusetts

Source: company disclosures and interviews conducted for this article.

Open Questions

Solvane's harder problem may not be chemistry or manufacturing but positioning. Kestrel Materials, Tanaka's former employer, has since started its own sodium-ion research effort in San Diego, and at least one larger incumbent has begun licensing similar cathode formulations for pilot lines. Ferreira does not dispute that the competitive field is filling in. "Two years ago we had this to ourselves," she said. "Now the question is whether we can out-execute people with a lot more capital."

For now, the Reno warehouse floor keeps stacking cells, and the order book, according to Ferreira, extends into the first half of next year. Whether that is enough runway to stay ahead of better-funded rivals is, by her own admission, the open question the company is building toward answering.